

packages/graph-view/src/utils/elk-layout.ts

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Overview

A new utility `packages/graph-view/src/utils/elk-layout.ts` adds ELK-specific types and a repair pipeline that sanitizes graph input before layout. The file imports `GraphIssue` from `@novadiff/graph-core/schema`, `loadElk` from `./elk-bundled`, and default node dimensions from `./layout` (lines 1-3).

Key changes

- **Types** (`lines 5-34`, `39-43`, `214-221`): `ElkChild`, `ElkEdge`, `ElkInput`, `RepairResult`, `ElkLayoutOptions`, `ElkLayoutResult`.
- **Repair logic** (`repairElkInput`, `lines 56-213`):
 - Fills missing `width / height` with defaults.
 - Deduplicates child IDs per parent.
 - Drops orphan children, orphan edges, and containment cycles.
 - Emits `GraphIssue` objects and may throw when `strict` is true.
- **Layout wrapper** (`applyElkLayout`, `lines 223-243`):
 - Calls `repairElkInput`, loads ELK via `loadElk`, runs `elk.layout`, and returns the positioned graph plus any issues.
 - On layout failure, returns a fatal `GraphIssue` unless `strict` is set.

Impact

- **Correctness**: Input is sanitized before ELK runs, reducing runtime errors.
- **Observability**: Issues are surfaced as `GraphIssue` objects, aiding debugging.
- **Compatibility**: Existing callers are unaffected; new functions are exported for optional use.

Risks & follow-ups

- Verify that `loadElk` resolves correctly in all target environments; a missing bundle could silently fail.
- Confirm that the `strict` flag propagates errors as intended.
- Test edge cases: graphs with many duplicate IDs, deep containment cycles, and orphan references.
- Monitor memory usage for large graphs; recursive traversal may be costly.